

2 (short) A general voltage amplifier is shown in Fig. 2.

- (a) For the case where the output is an open-circuit, determine the voltage gain. [1]
- (b) A load resistor, R , is connected across the output terminals BC. Determine the new voltage gain of the resulting circuit. [3]
- (c) For the case that the source of v_{in} has zero internal resistance and the value of $R = R_{out}$, determine the power gain of the circuit. [4]
- (d) Show that the condition for maximum transfer of power to the load is that $R = R_{out}$. [2]

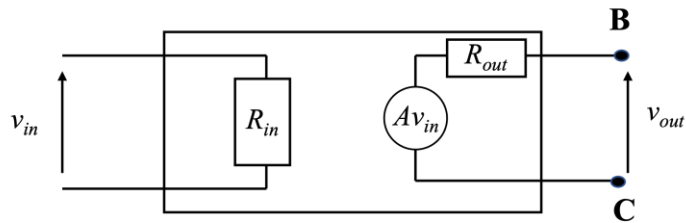


Fig. 2